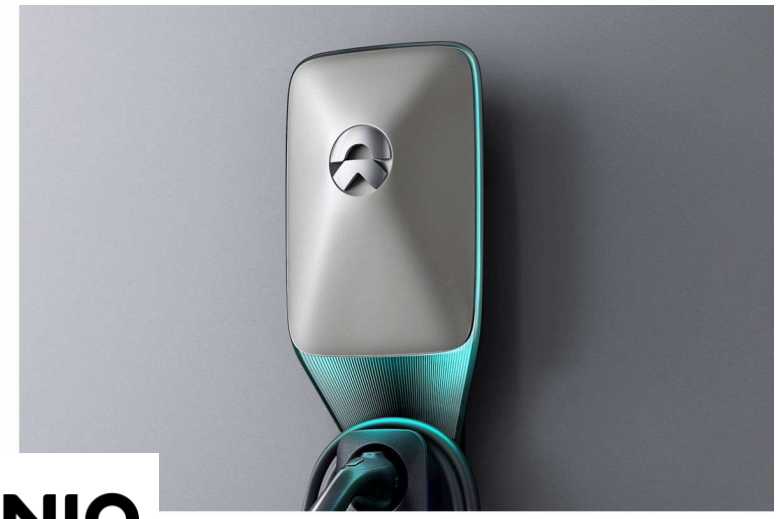
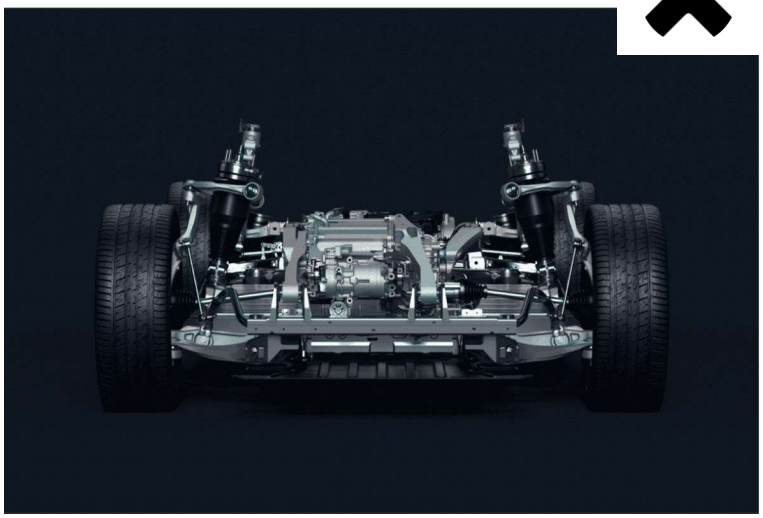
Student Hackathon

Pre-Registration Information





We are NIO.
The Next Generation Car Company

NIO is a global electric-vehicle and technology company, working at the intersection of engineering, software, and AI-driven mobility.

Project Background

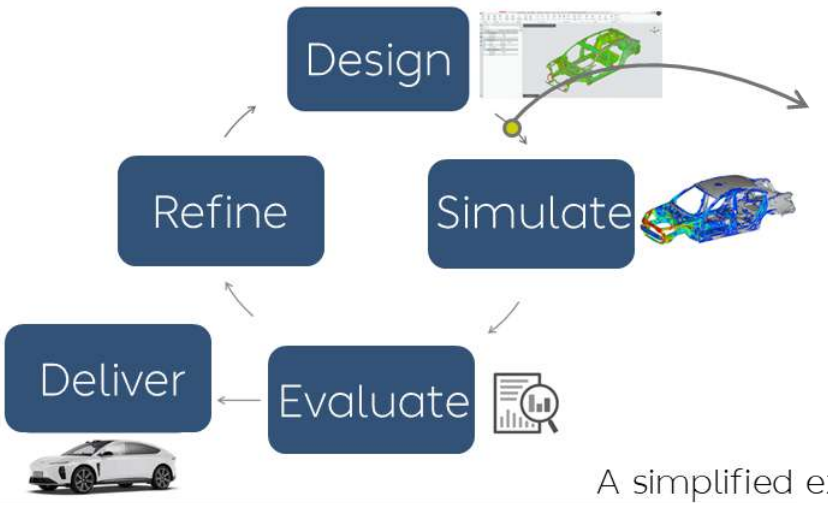
In NIO's automotive engineering department, an iterative process is used to refine the Computer Aided Design (CAD) models. This iterative engineering process (outlined below) includes various simulated tests, i.e. crash testing.

These simulations rely on very high quality CAD models to produce accurate results. Therefore, a series of automated Quality Assurance (QA) checks are performed between the 'Design' and Simulate' steps to ensure the inputs to the simulation software meet the required quality standards.

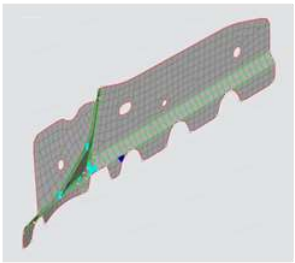
These QA checks are performed at the part level, an example of which is below.

Several checks are performed on each part, and each check requires bespoke scripts to be developed. For example, a check is in place to ensure holes for bolts and fixings are meshed correctly. Details of this current computer vision method to check meshed holes is outlined below.

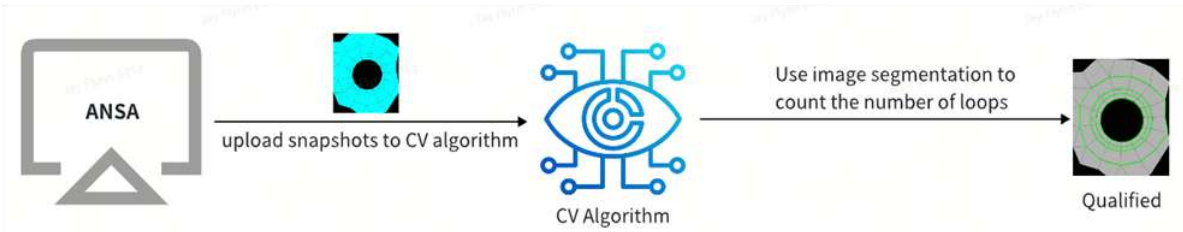
The hackathon contributes towards the development of an improved method to assess the quality of meshed holes.



QA checks are performed to ensure mesh objects meet the required quality standards for simulation.



A simplified example of a QA approach to check the quality of mesh holes.



Wider Research Context

The proposed improved method for assessing the quality of meshed holes explores the use of convolutional neural networks (CNNs) trained as a binary classifier to distinguish “Good” and “Bad” mesh quality. This method offers a more generalised and scalable approach that can also be replicated across other QA tasks.

At present, the primary limitation to adopting CNN-based methods is the lack of sufficiently large, labelled training datasets, particularly for failure cases.

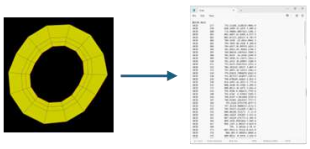
To address this, the methodology outlined below proposes a synthetic data generation pipeline capable of producing very large volumes of labelled training images.

The hackathon focuses on Step 1 of the proposed methodology: processing .jpeg images to extract relevant geometric information, and exporting in a defined file format.

Further technical details, including data formats, evaluation criteria, and example inputs, will be released to participants ahead of the event.

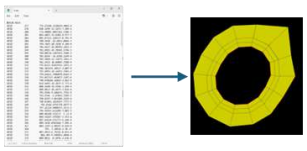
1 .jpeg to .nas file conversion (Hackathon Topic)

Import .jpeg images → Pre-process images → Extract Node Coordinates → Identify element Nodes → Output nodes & elements in .nas format



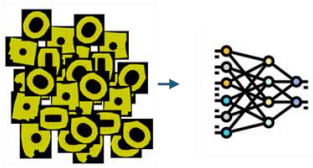
2 .nas file manipulation to generation high volume training dataset

Import .nas file → Modify mesh to augment image → Reproduce this process to create very large dataset



3 Train CNN, Multi-modal LLM , or similar approaches, with new expanded dataset

Train using synthetic images → Test using original images → Compare range of CNNs & other methods → Evaluate performance



4 Explore opportunities to replicate this approach in other QA checks.

Competition Details

Registration opens one week prior to the event, for teams of 1 to 4 students.

The competition runs virtually over ~1 month, during which teams work independently on a Python-based image processing challenge using datasets released at the start of the event.

A £4,000 cash prize will be awarded and split evenly among the winning team.

Beyond the prize, the hackathon provides hands-on experience of a real-world engineering and data analytics challenge.

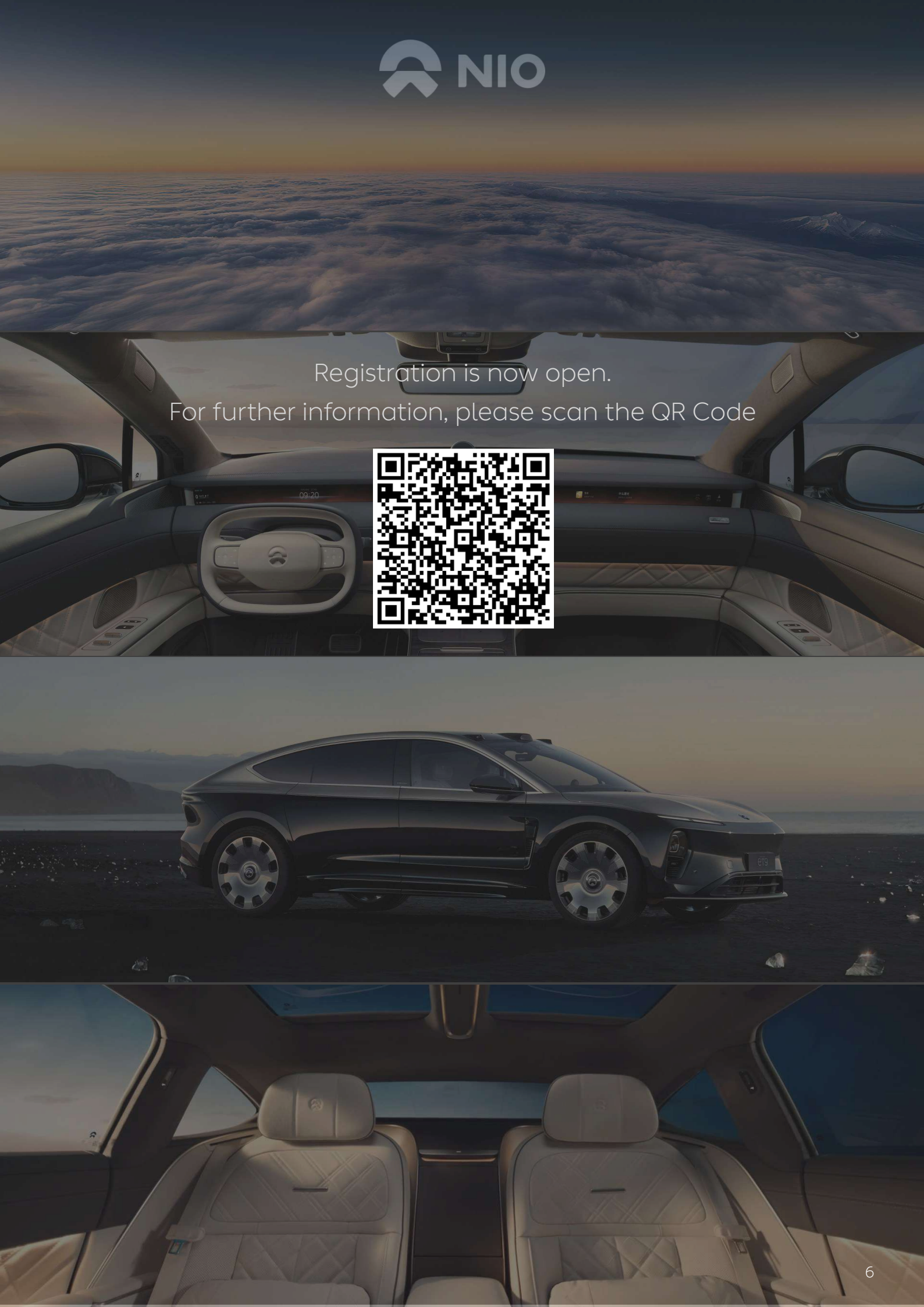
As part of a wider talent-scouting initiative, there will also be an opportunity for the winning team to co-author a conference paper with NIO on the research topic outlined on page 4, with further opportunities for academic collaborations including MSc, MRes, YII, and PhD project opportunities.

Entry is open to any university student currently enrolled on a UK university course of study.

The competition will run from **23rd of March — 17th April**. To participate, all members of your team must sign up using the link below.

Registration is now open

Please scan the QR code below to register for the event.


NIO
Registration is now open.
For further information, please scan the QR Code